

Algebra First-Year Exam, January 2025, Part 1

Answer 4 questions. You should indicate which 4 problems you wish to be graded. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

1. In S_8 , write down a representative of each conjugacy class of elements of order 4 and compute the order of its centralizer.
2. Let G be a group of order $124 = 2^2 \cdot 31$. Prove that G has a cyclic subgroup of order 62.
3. Prove that there exists a nonabelian group of order $2025 = 3^4 \cdot 5^2$. (You do not have to determine its multiplication explicitly.)
4. This problem has three parts.
 - (a) Give the definition of a maximal subgroup of a nontrivial group.
 - (b) Find two maximal subgroups of S_5 of different orders.
 - (c) Show that the group $(\mathbb{Q}, +)$ has no maximal subgroup.
5. Let G be a finite group and p a prime. Let K be the intersection of all the Sylow p -subgroups of G .
 - (a) Prove that K is a characteristic subgroup of G .
 - (b) Suppose H is a normal p -subgroup of G . Prove that $H \subseteq K$.
6. Let G be a group, and let $Z(G)$ denote its center.
 - (a) State what it means for G to be nilpotent of class c .
 - (b) Prove that if G is nilpotent of class 1, then G is abelian.
 - (c) Define the commutator subgroup G' of G .
 - (d) Prove that if G is nilpotent of class 2, then $G' \subseteq Z(G)$.